SASITHARAN M.S

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GITHUB sasitharan06

EDUCATION

B.Tech - AI & DS	Sri Eshwar College of Engineering	**CGPA: 8.19 (Upto 3rd semester)**	**2024-2028**
HSC	Thiagarajar Higher Secondary School	**92.5 %**	**2022-2024**
SSLC	Thiagarajar Higher Secondary School	**89.6 %**	**2021-2022**

PROJECTS

TRACKIT – Expiry Tracking & Predictive Analysis System

Tech stack: Next.js, MongoDB, Pandas, Matplotlib, OpenCV, Tesseract OCR.

Built a full-stack web application for real-time expiry tracking of packed and medical products using barcode scanning & OCR-based label extraction. Designed the frontend with Next.js and managed customer and product data using MongoDB. The system links each purchase to the customer's contact, enabling personalized expiry alerts and return facilitation.

Integrated OpenCV and Tesseract OCR for accurately extracting expiry dates from product packaging. Used Pandas and Matplotlib to analyze expiry trends and provide predictive insights for sellers on restocking and product movement. Added a chatbot feature to assist customers with expiry queries, return policies, and product safety information, enhancing user experience

Live Face Attendance Tracking System using AI

Tech stack: Python, OpenCV, Face Recognition, NumPy, Pandas, SQLite.

Developed a real-time AI-based face attendance tracking system using Python to automate and secure the process of attendance logging in classrooms and offices. Integrated OpenCV and a pre-trained face recognition model to detect and recognize multiple faces from a live webcam feed with high accuracy.Used NumPy and Pandas to process attendance logs and generate reports. The system stores user data and attendance records in a lightweight SQLite database for easy access and management. Implemented a user-friendly GUI for admin control, new user registration, and attendance visualization.

The system ensures spoof prevention by checking for live face detection cues and allows exporting of daily/monthly attendance reports, streamlining workforce and academic attendance management.

Smart Waste Segregation System using AI

Tech stack: Python, OpenCV, TensorFlow, Scikit-learn, Streamlit, NumPy, Pandas

Developed an AI-powered waste segregation system to automatically classify solid waste into categories such as plastic, metal, and organic using image processing and machine learning. Utilized OpenCV for image preprocessing and a trained deep learning model built with TensorFlow and Scikit-learn for real-time waste classification.

Integrated the system into a Streamlit-based web interface for interactive visualization and testing. Implemented data handling and model evaluation using NumPy and Pandas to ensure high accuracy and efficiency in prediction.

The system promotes environmental sustainability by enabling automated waste management and segregation, reducing human effort, and enhancing recycling processes.

CERTIFICATIONS

Machine learning course(NSDC)- Link	**2025**
Basics in C (great learning)- Link	**2024**
SQL(basis) - Link	**2025**

CODING PROFILES

Leetcode | **250+** solved | **Global rank 251,662** in Leetcode | **Rating** : **1557** | Profile Link

Hackerrank | 2 Star C++ , 3 Star C,4 star MYSQL | Profile Link

Skillrack | **Solved 1300+** Problems | Profile Link

ACHIEVEMENTS

Top 50 in Skillrack college level – 2024

National level non-tech events runners in PSG Tech College -2024

High Cutoff in HSC top 3 In my school level

TECHNICAL SKILLS

Programming - (Basis) Python | C++ | C | HTML | CSS | MYSQL | Data Structures | JAVA

Artificial Intelligence - Machine Learning

Full Stack : MERN Stack

Tools : Pycharm | Visual Studio Code | Git | Github

Backend Software Development -Streamlit | MongoDB